



Towards Zero-Shot Annotation of the Built Environment with Vision-Language Models

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Abstract

Equitable urban transportation applications require high-fidelity digital representations of the built environment (streets, crossings, curb ramps and more). Direct inspections and manual annotations are costly at scale, while conventional machine learning methods require substantial annotated training data for adequate performance. This study explores vision language models as a tool for annotating diverse urban features from satellite images, reducing the dependence on human annotation. Although these models excel at describing common objects in human-centric images, their training sets may lack signals for esoteric built environment features, making their performance uncertain. We demonstrate a proof-of-concept using a vision language model and a visual prompting strategy that considers segmented image elements. Experiments on two urban features — stop lines and raised tables — show that while zero-shot prompting rarely works, the segmentation and visual prompting strategies achieve nearly 40% intersection-over-union accuracy. We describe how these results motivate further research in automatic annotation of the built environment to improve equity, accessibility, and safety at scale and in diverse environments.

CCS Concepts

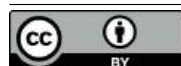
• **Applied computing**; • **Computing methodologies** → **Scene understanding**; *Visual inspection*; **Image segmentation**;

Keywords

Large Language Model, Vision Language Model, Image Segmentation, Urban Data Annotation, Urban Computing

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1 Introduction

The inaccessible urban infrastructure reinforces systemic exclusion of people with disabilities and negatively impacts public health and overall quality of life for everyone [5]. As laws evolve to make the urban environment more accessible, the digital representation must also evolve to include not just roads and sidewalks, but the hundreds of features that enable safe and equitable mobility: curb ramps, walk signals, tactile paving, crosswalks, light poles, benches, and many more [12]. For example, mobility applications that provide accessible pedestrian routes rely on accurate, comprehensive annotations of the dynamic urban infrastructure [2].

The gold standard for data collection involves in-person inspections, typically administered by local governments with the help of community volunteers or contractors [6]. Human annotations are high quality but are labor-intensive, costly, and difficult to scale. For instance, in 2017, the city of Seattle employed 14 people for their first-ever audit of 2,300 miles of sidewalks, costing \$400,000 and identifying 92,000 uplifts, 38,000 surface problems, and 20,000 obstructions [6]. Sensing platforms deployed to navigate the built environment can partially automate annotation via machine learning, but these initiatives are also expensive to scale [1]. Aerial imagery offers nearly complete coverage at low cost, as long as the computer vision models to extract the relevant features can be trained. In practice, model development tends to focus on the needs of the "typical" traveler, e.g., roads [3] and in a few cases sidewalks [21]. Expanding coverage to the long tail of accessibility features is rarely viable due to the cost in obtaining training data.

Pre-trained vision-language models (VLMs) (e.g., [11, 13]), exhibit impressive multi-modal scene understanding that can be used, among many other tasks, to identify objects in images [14]. If these models could be used to bootstrap feature annotation in the built environment, a more complete digital record could be developed enabling equity and accessibility at scale [7]. However, these models are trained on text paired with images from the internet, such that common objects in everyday (US-centric) experiences tend to be dominant. For atypical scenes and images, including satellite images, VLMs' performance on them remains to be tested.

In this paper, we consider using VLMs with aerial imagery as a source of low-cost, extensible annotations for arbitrary features in the built environment. By reducing the cost of producing annotations to near zero, mobility applications can keep pace with an evolving building code and commensurate new construction,

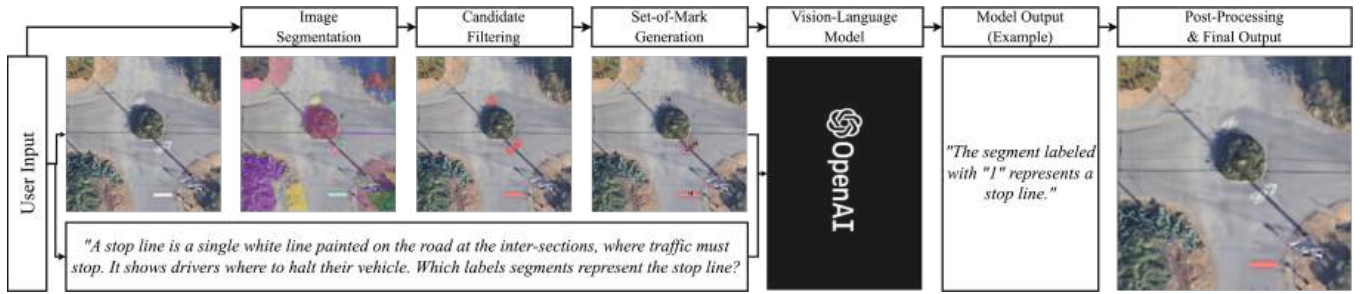


Figure 1: Pipeline of our proposed automated annotation process. Users input a pair of (satellite image, annotation guidance). The image will go through a set of processes including segmentation, filtering, and set-of-mark generation. Then the image and guidance will go through a vision-language model, the output of which is post-processed to produce the final annotation results. The procedure requires no fine-tuning, and can be applied on different features with minimal adjust on the guidance.

faithfully representing the complete and current built environment. With this capability, we envision a larger road-map to scale this annotation process to any feature in any region, dramatically lowering costs for data acquisition and therefore enable a broad class of community-focused, AI-based mobility applications that have previously been too specialized to justify the cost.

To demonstrate the proof-of-concept, we develop a procedure borrowing techniques from visual grounding [18, 22] and visual prompting [15, 17] that generates bounding box annotations on satellite images based only on high-level descriptions of features. The procedure requires no fine-tuning or even labeled examples, demonstrating that closed, general purpose models are potentially viable even in a zero-shot setting. We use the a state-of-the-art (SOTA) VLM (GPT-4o¹) and a general-purpose segmentation model (SAM [8]). The SAM model segments a satellite image into a gallery of candidate objects, from which the GPT-4o model can select the appropriate feature of interest (see Figure 1 for the procedure).

We test our procedure on two urban features: stop lines (white lines indicating where cars should stop at an intersection, which are important for safely deriving pedestrian routes through unmarked crossings) and raised tables (speed control features that can be used to estimate traffic flow and therefore pedestrian risk.) We consider multiple variants of this procedure, showing that presenting the segmented “options” to the model in different ways can improve performance. Quantitatively, we show that a simple direct prompting strategy fails dramatically at this task, while the proposed procedures achieve on average about 40% intersection-over-union (IoU) accuracy. The takeaway is that this problem is difficult, but not impossible. Qualitatively, we visualize several correct and incorrect annotations to inform a discussion of the challenges and vision. Overall, we make the following contributions:

- We describe a prompting procedure for generating annotations urban features from aerial imagery using a VLM, adapting segmentation based annotations methods to this context.
- We evaluate the procedure on two urban features, finding that while simple methods fail, the proposed methods are promising and viable, which suggests an important avenue for research.
- Given these results, we discuss the current challenges faced by the automated process and describe how a complete, accurate, and publicly available data record may now be feasible and consider next steps for the research community to realize this vision.

¹<https://platform.openai.com/docs/models/gpt-4o>

2 Related Work

Urban Data Collection: Zhang et al. [21] introduced a novel dataset comprising aerial satellite imagery, street maps, and rasterized annotations of street objects in cities. Even after years of work, existing models recognize only the three most important features of the pedestrian network. These models require expansive, diverse training data to perform well; our focus is on a process for collecting this training data for rare cases and at large scale.

Visual Prompting: Visual prompting is a technique that uses visual cues or examples to guide multi-modal AI models in understanding and responding to visual tasks, similar to how textual prompts are used to instruct LLMs [15]. Our proposed procedure is based on Set-of-Mark (SoM) prompting [17], leveraging segmentation model SAM to partition images into smaller regions, and overlays each region with a mark to provide visual hints. Their work demonstrated extraordinary visual grounding ability of GPT-4.

Visual Grounding (VG): our problem aligns with the VG task, which involves annotating objects in images by providing textual prompts to VLMs [18, 22]. However, VG tasks are typically evaluated on standard images (e.g., sightseeing, animals, buildings). To our knowledge, only a few studies have utilized VLMs to annotate objects in aerial images. Studies [4, 20] annotated generic objects from aerial images across a wide range of categories, such as basketball courts, harbors, and stadiums. They trained small VLMs from scratch, instead of using pre-trained large-scale VLMs. Studies [10, 19] created benchmark datasets for annotation, which only include commonly seen features such as roads, cars, and buses. Our study is different from them in that we focus on the scalability, generalizability, and cost-effectiveness of annotating diverse urban features, instead of just common objects. We leverage the SOTA VLM, generic segmentation model (SAM), and visual prompting strategy to enhance annotation performance, without requiring any labeled training data.

3 Prompting Procedure

Our proposed procedure is shown in Figure 1. It works as follows:

- **User Input:** a pair of (satellite image, annotation guidance).
- **Image Segmentation:** The image is segmented into a set of candidates using a general-purpose model, ignoring the prompt.
- **Candidate Filtering:** We apply heuristic filters to eliminate irrelevant components and narrow down the candidate space. In our experiments, we employ color and area size filters, excluding

green, yellow, brown, and red elements, as well as components smaller than specific sizes². In practice, different parameter values and additional filter criteria can be applied based on the specific urban features and expert knowledge.

- **Set-of-Mark (SoM) Generation:** After filtering, we generate an identifying *mark* for each component [17] to enhance the vision-language model’s recognition of each object. We test three different marking strategies as shown in Figure 2:
 - **No-Context (NC):** the candidates are extracted and presented as distinct, unrelated objects embedded in a plain white background, with each one labeled with a number (Figure 2(b)).
 - **In-Context (IC):** we label the candidates with numbers and bounding boxes within the original image (Figure 2(c)).
 - **Combination (Comb):** We generate both no- and in-context images and provide both to the VLM.
- **VLM & Post-Processing:** the image from the SoM generation process is sent to the VLM with the textual annotation guidance. The text prompt consists of 1) a description of the feature of interest (e.g., stop line, raised table) and what specific features it possesses and 2) the instructions to annotate the object (see Table 1 for stop line example). The output of the model is post-processed: the model responds with the mark indicating the selected candidate and corresponding bounding box is returned.

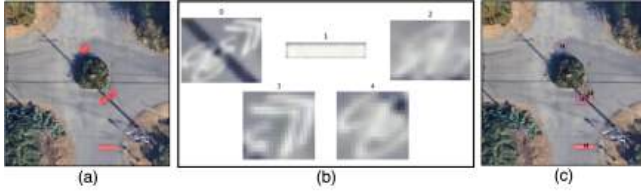


Figure 2: (a) Filtered segmented candidates. (b) No-Context — candidate objects are presented separately in a new image. (c) In-Context — candidate objects are labeled with numbers and bounding boxes within the original image.

Method	Stop Line
Direct Prompting	A stop line is a single white line painted on the road at intersections where traffic must stop. It shows drivers where to halt their vehicles. Please identify the bounding box of the stop line in the image in the format of (x1, y1, x2, y2).
SoM Prompting	A stop line is a single white line painted on the road at intersections where traffic must stop. It shows drivers where to halt their vehicles. Which labeled images represent the stop line?

Table 1: Example text prompts for the VLM model to annotate stop lines. Pink highlighted texts represent the descriptive information of the features. Blue highlighted texts represent the general task questions.

4 Evaluation & Results

To evaluate this procedure, we manually cropped and annotated eight satellite images that contains stop lines and ten images that contain raised tables from Google Earth. As a baseline, we compare against a **Direct Prompting (DP)** method: given an image, we directly ask GPT-4o (without SAM) to annotate the feature of interest and return the bounding boxes. Quantitatively, we evaluate the annotations using Intersection-over-Union (IoU) metric: $IoU = \frac{G \cap P}{G \cup P}$.

²200 pixel area for stop lines and 400 pixel area for raised tables.

where G is the ground truth annotation and P is the annotation from our procedure. The results are presented in Table 2.

We observe that — 1) Direct prompting is completely ineffective, achieving essentially zero overlap with ground truth. The annotations reflect very little understanding of either the instructions or the image, which is consistent with prior work [17]. 2) Adding visual marks significantly improves performance, achieving 24%-40% IoU depending on the variant. In-context improves over no-context by 47.3% and 26.4% on two features respectively, and providing both forms yields an additional slight improvement.

Qualitatively, we visualize a few annotated examples to demonstrate the feasibility of the automated process. We also mark some flawed examples to be discussed in the next section §5. From the visualizations, we observe that — 1) the failure of direct prompting, indicating the problem is difficult and not amenable to trivial solutions; 2) given visual marks, the VLM model can always at least partially identify (highlighted with yellow boxes), and occasionally fully identify the features (highlighted with green boxes), suggesting the problem is feasible for other types of features; and 3) there is no consistent winner among the three annotation strategies.

Feature	DP	SoM-NC	SoM-IC	SoM-Comb
Stop Line	0.0000	0.2483	0.3354	0.3657
Raised Table	0.0190	0.3315	0.4069	0.4189

Table 2: Quantitative evaluation results using IoU(█) metric.

5 Challenges & Future Direction

Manually annotating urban features is labor-intensive and time-consuming. Our proposed procedure reduces the annotation process to — a) describing a typical representation of the feature, and 2) providing simple heuristics for filtering candidates based on color, size, and other basic properties. We evaluate in the zero-shot setting to reduce the dependency on finding representative examples — a user can directly apply domain knowledge to rapidly annotate any feature data from satellite images. Our experiments also expose several limitations and motivate directions for further work:

- **Noisy/Missing Segmentation.** The annotation results heavily depend on the quality of the segmented components — SAM might miss part of or the entire feature. For example, part of the raised table is covered by the shadow, thus not being detected; or SAM could possibly not recognize rare urban features at all. To address the issues, we recommend studying segmentation models specialized for satellite images that can, e.g., ignore shadows.
- **Domain Expertise & Partial Manual Process is Still Required.** Since SAM is a general-purposed segmentation model, it often produces irrelevant segments. Currently, we prune the space of candidates heuristically to improve performance, necessitating some manual process. Additionally, these heuristics may need to be developed for each feature individually, which requires some domain expertise to achieve desired results. It is unclear how to perform this task in a principled way yet.
- **Capability of Vision-Language Models.** The rapid improvements in VLMs mediate our results — 1) Anecdotal, the recently released GPT-4o dramatically outperformed the state of the art open VLM and yet still makes mistakes, suggesting limited comprehension of the built environment. For example, crossing lines



Figure 3: Left – Examples of annotated stop lines. Right – Examples of annotated raised tables. Green and yellow outlines indicate perfect and approximate annotations, respectively. Red outlines indicate inaccurate annotations.

oriented parallel to the roadway are mistaken for stop lines (Figure 3, example 5, SoM-NC, yellow box) or misunderstand the task entirely (Figure 3, example 1, SoM-NC, red box). VLMs fine-tuned on satellite images may avoid these errors. 2) We only evaluated zero-shot prompting in this study (e.g., no examples provided). Considering that VLMs/LLMs are sensitive to prompting techniques, we recommend evaluating more advanced prompting structures, such as few-shot (compatible with LLaVA-Next[9] and GPT-4o) and chain-of-thought prompting [16] etc.

- **City-wise Generalizability.** Since the process relies on two general-purpose large AI models, it can potentially generalize to any feature in any location. However, this generalization remains untested. Some features of interest will be significantly harder to recognize: potholes have irregular shapes and sizes; benches will be more often shaded or occluded. Further, laws and standards vary across cities, changing the visible environment.

References

- [1] Andrew R Benson, Ian GM Lawson, Matthew K Clifford, and Sean M McBride. 2021. Using robotics to detect footpath displacement caused by tree roots: A proof of concept. *Urban Forestry & Urban Greening* 65 (2021), 127312.
- [2] Nicholas Bolten and Anat Caspi. 2019. AccessMap Website Demonstration: Individualized, Accessible Pedestrian Trip Planning at Scale. 676–678. <https://doi.org/10.1145/3308561.3354598>
- [3] Luca Caltagirone, Mauro Bellone, Lennart Svensson, and Mattias Wahde. 2019. LIDAR–camera fusion for road detection using fully convolutional neural networks. *Robotics and Autonomous Systems* 111 (2019), 125–131.
- [4] Yueli Ding, Haojie Xu, Di Wang, Ke Li, and Yumin Tian. . Visual Selection and Multi-Stage Reasoning for RSVG. *IEEE Geoscience and Remote Sensing Letters* ().
- [5] Yochai Eisenberg, Kerri A Vanderbom, and Vijay Vasudevan. 2017. Does the built environment moderate the relationship between having a disability and lower levels of physical activity? A systematic review. *Preventive medicine* 95 (2017).
- [6] Jon E Froehlich, Yochai Eisenberg, Maryam Hosseini, Fabio Miranda, Marc Adams, Anat Caspi, Holger Dieterich, Heather Feldner, Aldo Gonzalez, Claudina De Gyves, et al. 2022. The Future of Urban Accessibility for people with disabilities: Data collection, analytics, policy, and Tools. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–8.
- [7] Bill Howe, Jackson Maxfield Brown, Bin Han, Bernease Herman, Nic Weber, An Yan, Sean Yang, and Yiwei Yang. 2022. Integrative urban AI to expand coverage, access, and equity of urban data. *The European Physical Journal Special Topics* 231, 9 (2022), 1741–1752.
- [8] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4015–4026.
- [9] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. 2024. LLaVA-OneVision: Easy Visual Task Transfer. *arXiv preprint arXiv:2408.03326* (2024).
- [10] Xiang Li, Jian Ding, and Mohamed Elhoseiny. 2024. VRSBench: A Versatile Vision-Language Benchmark Dataset for Remote Sensing Image Understanding. *arXiv preprint arXiv:2406.12384* (2024).
- [11] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2024. Visual instruction tuning. *Advances in neural information processing systems* 36 (2024).
- [12] CHRISTINE MENDOZA and ANAT CASPI. 2023. Development and Taxonomization of an Accessibility-First Open Data GIS Standard. University of North Carolina at Chapel Hill and University of Washington. <https://christineym.github.io/dreu-2023/files/finalreport.pdf>
- [13] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PMLR, 8748–8763.
- [14] Aleksandar Shtedritski, Christian Rupprecht, and Andrea Vedaldi. 2023. What does clip know about a red circle? visual prompt engineering for vlms. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 11987–11997.
- [15] Aleksandar Shtedritski, Christian Rupprecht, and Andrea Vedaldi. 2023. What does clip know about a red circle? visual prompt engineering for vlms. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 11987–11997.
- [16] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.
- [17] Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. 2023. Set-of-mark prompting unleashes extraordinary visual grounding in gpt-4v. *arXiv preprint arXiv:2310.11441* (2023).
- [18] Li Yang, Yan Xu, Chunfeng Yuan, Wei Liu, Bing Li, and Weiming Hu. 2022. Improving visual grounding with visual-linguistic verification and iterative reasoning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9499–9508.
- [19] Zhenghang Yuan, Lichao Mou, Yuansheng Hua, and Xiao Xiang Zhu. 2024. Rrsi: Referring remote sensing image segmentation. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
- [20] Yang Zhan, Zhitong Xiong, and Yuan Yuan. 2023. Rsvg: Exploring data and models for visual grounding on remote sensing data. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–13.
- [21] Yuxiang Zhang, Nicholas Bolten, Sachin Mehta, and Anat Caspi. 2023. APE: An Open and Shared Annotated Dataset for Learning Urban Pedestrian Path Networks. *arXiv preprint arXiv:2303.02323* (2023).
- [22] Chaoyang Zhu, Yiyi Zhou, Yunhang Shen, Gen Luo, Xingjia Pan, Mingbao Lin, Chao Chen, Lijuan Cao, Xiaoshuai Sun, and Rongrong Ji. 2022. Seqtr: A simple yet universal network for visual grounding. In *European Conference on Computer Vision*. Springer, 598–615.